

Architecture Selection:

Self-Service Coffee Kiosk System

Selected Architecture: Layered Architecture

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Reason 1: Layered architecture separates the presentation, business logic, and data storage layers. The presentation layer handles touchscreen interactions, while the business layer manages coffee orders and payments.

Reason 2: This architecture supports the kiosk's requirements, including menu and pricing storage, credit-card payments, and receipt printing. Each component has a clear responsibility, making the system easier to understand and maintain.

Security Advantage: Separating the user interface from the business logic helps restrict payment processing to the payment service component and reduces direct access to stored data.

Performance Benefit: The separation of responsibilities allows each component to perform its specific task efficiently, supporting the ordering and payment workflow.